STAT 6410
Homework 7

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1. **Chapter 3, Problem 8**

   Let X be a nonnegative random variable on some probability space.

   (a) Show that $(EX)(E\frac{1}{X}) \geq 1$. What does this say about the correlation between X and $\frac{1}{X}$.

   (b) Let $f, g : \mathbb{R}_+ \to \mathbb{R}_-$ be Borel measurable and such that $f(x)g(x) \geq 1$ for all $x \in \mathbb{R}_+$. Show that $Ef(X)Eg(X) \geq 1$.

   (You can assume $X > 0\, a.e.(P)$, but it is not needed)

Ans: (a)

Proof. Suppose X is a nonnegative random variable on some probability space.

Case 1: $E(X) < \infty$

Case A: $E(\frac{1}{X}) < \infty$

$1 = EX\frac{1}{EX} \leq EXE(\frac{1}{X})$, by Jensen's Inequality since $g(x) = \frac{1}{x}$ is a convex function.

Case B: $E(\frac{1}{X}) = \infty$

$\infty \geq E(\frac{1}{X}) \geq \frac{1}{E(X)}$ holds.

$1 = EX\frac{1}{EX} \leq EXE(\frac{1}{X})$

Case 2: $E(X) = \infty$

Thus, $\frac{1}{E(X)} = 0$. Since $X \geq 0$, $E(\frac{1}{X}) \geq 0$. Thus, $0 = \frac{1}{E(X)} \leq E(\frac{1}{X})$

$1 = EX\frac{1}{EX} \leq EXE(\frac{1}{X})$ $\square$

This is the formula for the correlation between X and $\frac{1}{X}$:

$$Corr(X, \tfrac{1}{X}) = \frac{E(X\frac{1}{X}) - (EX)(E\frac{1}{X})}{\sqrt{Var(X)Var(\frac{1}{X})}} = \frac{1 - (EX)(E\frac{1}{X})}{\sqrt{Var(X)Var(\frac{1}{X})}}$$

Clearly, $\sqrt{Var(X)Var(\frac{1}{X})} \geq 0$, so $Corr(X, \frac{1}{X}) \leq 0$.

(b)

Proof. $X > 0$, so $f(X)g(X) \leq 1$. Thus, $g(X) \leq \frac{1}{f(X)}$.
Therefore, $Ef(X)Eg(X) \leq Ef(X)E\frac{1}{f(X)} \leq 1$, from part (a). □

2. **Chapter 3, Problem 9**

   Prove Corollary 3.1.13 using Hölder's inequality applied to an appropriate measure space.

Ans:

Proof. Define $(\Omega, \mathcal{F}, \mu)$ to be measure space such that $\Omega = \mathbb{N}$, $\mathcal{F} = \sigma\langle\{\{i\}\}_{1 \leq i \leq k}\rangle$, and μ is the counting measure.

Let $k \in \mathbb{N}$. Let $a_1, a_2, ..., a_k, b_1, b_2, ..., b_k \in \mathbb{R}$ and $c_1, c_2, ..., c_k \in \mathbb{R}_+$.

Part (a):

Define: $f \equiv \sum_{i=1}^{k} c_i^{1/p}|a_i|I(\{i\}) = \sum_{i=1}^{k} c_i^{1/p}|a_i|$
and $g \equiv \sum_{i=1}^{k} c_i^{1/q}|b_i|I(\{i\}) = \sum_{i=1}^{k} c_i^{1/q}|b_i|$.
Thus, $fg = \sum_{i=1}^{k} c_i|a_ib_i|I(\{i\}) = \sum_{i=1}^{k} c_i|a_ib_i|$
Let $1 < p < \infty$ and $q \equiv \frac{p-1}{p} \implies \frac{1}{q} + \frac{1}{p} = 1$

$$|fg| = \sum_{i=1}^{k} c_i|a_ib_i| = \sum_{i=1}^{k} c_i|a_ib_i|I(\{i\}) = \int c_i|a_ib_i|d\mu(\{i\})$$

$$= \int |fg|d\mu(\{i\}) \leq \left(\int |f|^p d\mu(\{i\})\right)^{1/p} \left(\int |q|^q d\mu(\{i\})\right)^{1/1} \quad \text{(By Hölder's Inequality)}$$

$$= \left(\int |a_ic_i^{1/p}|^p d\mu(\{i\})\right)^{1/p} \left(\int |b_ic_i^{1/q}|^q d\mu(\{i\})\right)^{1/q}$$

$$= \left(\sum_{i=1}^{k} |a_i|^p c_i\mu(\{i\})\right)^{1/p} \left(\sum_{i=1}^{k} |b_i|^q c_i\mu(\{i\})\right)^{1/q}$$

$$= \left(\sum_{i=1}^{k} |a_i|^p c_iI(\{i\})\right)^{1/p} \left(\sum_{i=1}^{k} |b_i|^q c_iI(\{i\})\right)^{1/q}$$

$$= \left(\sum_{i=1}^{k} |a_i|^p c_i\mu(\{i\})\right)^{1/p} \left(\sum_{i=1}^{k} |b_i|^q c_i\mu(\{i\})\right)^{1/q}$$

$$= \left(\sum_{i=1}^{k} |a_i|^p c_i\right)^{1/p} \left(\sum_{i=1}^{k} |b_i|^q c_i\right)^{1/q}$$

Thus, $\sum_{i=1}^{k} |a_i b_i| c_i \leq \left(\sum_{i=1}^{k} |a_i|^p c_i\right)^{1/p} \left(\sum_{i=1}^{k} |b_i|^q c_i\right)^{1/q}$. $\square$

Part (b): Define $p = 2$. Thus, $q = 2$.

From Part (a), $\sum_{i=1}^{k} |a_i b_i| c_i \leq \left(\sum_{i=1}^{k} |a_i|^2 c_i\right)^{1/2} \left(\sum_{i=1}^{k} |b_i|^2 c_i\right)^{1/2}$

3. **Chapter 3, Problem 17 (a) and (b)**

   Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mu)$. Recall that $Eh(X) = \int h(X) d\mu$ if $h(X) \in L^1(\Omega, \mathcal{F}, \mu)$.

   (a) Show that $(E|X|^{p_1}) \leq (E|X|^{p_2})^{p_1/p_2}$ for any $0 < p_1 < p_2 < \infty$.

   (b) Show that '=' holds in (a) iff $|X|$ is a constant a.e.(μ)

Ans: (a)

Proof. Let $0 < p_1 < p_2 < \infty$ be given. Define: $Z \equiv |X|^{p_1}$. Note that $|Z| = Z$. Define: $a \equiv \frac{p_2}{p_1}$ and $b \equiv \frac{a}{a-1}$

$$E|X|^{p_1} = EZ \leq (EZ^a)^{1/a}(E1^b)^{1/b} \qquad \text{(using Hölder's Inequality)}$$
$$= (E(|X|^{p_1})^{p_2/p_1})^{\frac{1}{p_2/p_1}} = (E|X|^{p_2})^{p_1/p_2}$$

$\square$

(b)

Proof. Let $0 < p_1 < p_2 < \infty$ be given.

From the textbook, we know that the equality holds iff, $|Z|^a = c|1|^b$ for some constant $c \in (0, \infty)$
$|Z|^a = c|1|^b \implies |X|^{p_2/p_1} = c \implies |X| = c^{p_1/p_2}$. c^{p_1/p_2} is a constant since p_1 and p_2 are fixed values. $\square$

4. **Chapter 3, Problem 18**

   Let X be a nonnegative random variable

   (a) Show that $E(X \log(X)) \geq (EX)(E \log X)$.

   (b) Show that $\sqrt{1 + (EX)^2} \leq E(\sqrt{1 + X^2}) \leq 1 + EX$.

Ans: (a)

Proof. Define $g(x) = x\log(x)$. Note that $g(x)$ is convex for $x > 0$.
Thus, $E(h(X)) \leq h(EX) \implies EX\log(EX) \leq E(X\log X)$.

Define $h(x) = \log(x)$. Note that $h(x)$ is concave for $x > 0$.
Thus, $E(h(X)) \leq h(EX) \implies E(\log X) \leq \log(EX)$

Putting this all together, we get that:

$$E(X\log X) \geq EX\log(EX) \geq EXE(\log X)$$

□

(b)

Proof. Define $g(x) = \sqrt{1+x^2}$ and $h(x) = 1 + x$.
Note that for all $x \geq 0$, $g(x) \leq h(x)$ (*). Also note that both $g(x)$ and $h(x)$ are convex functions where $x > 0$.

From Jensen's Inequality, we know that: $\sqrt{1+(EX)^2} \leq E(\sqrt{1+X^2})$.

From (*), we know that $E(g(X)) \leq E(h(X)) \implies E(\sqrt{1+X^2}) \leq E(1+X) = 1 + EX$.

Putting this all together: $\sqrt{1+(EX)^2} \leq E(\sqrt{1+X^2}) \leq 1 + EX$. □